Paluri Sai Sasidhar

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Education

Indian Institute of Information Technology & Management, Gwalior **June 2023 – Expected May 2027**
Pursuing B.Tech in Computer Science and Engineering *Gwalior, Madhya Pradesh*

Relevant Coursework

Data Structures & Algorithms, Operating Systems, Database Management (DBMS), Computer Networks, Object-Oriented Programming, Cryptography & Network Security, Software Engineering, Design and Analysis of Algorithms.

Experience

Tecdatum Infoservices Pvt. Ltd. **May 2026 – July 2026**
AI Intern *Hyderabad, Telangana*

- **Edge AI & ANPR System:** Engineered a modular Python-based software architecture and Docker-based containerized deployment pipeline for an Automatic Number Plate Recognition (ANPR) system targeting industrial edge devices (NVIDIA Jetson Nano).
- **Hardware-Accelerated Interface:** Built a custom TensorRT inference pipeline to compile and execute YOLOv8 models, programmatically managing GPU memory allocation and asynchronous execution contexts to reduce latency by 68% while maintaining 95% accuracy.
- **Escape Route Optimization:** Designed a geospatial routing engine using OSMnx and NetworkX to model complex street grids; optimized pathfinding efficiency by developing a custom pruned Dijkstra algorithm that truncated search branches using spatial coordinates.

Projects

Triton Real-Time Engine | High-Throughput AI Inference & Quantization **August 2026**
C++, Python, PyTorch, ONNX, TensorRT, CUDA, Triton Inference Server, FastAPI, gRPC, Docker

- Engineered a high-performance C++/Python AI model-serving framework leveraging TensorRT and ONNX IR to deploy quantized deep-learning models (INT8/FP16) with 3.2× lower latency (80 ms → 25 ms).
- Architected thread-safe dynamic-batching asynchronous request queues in C++ and asyncio, boosting GPU compute utilization by 55% and scaling server throughput to 1,200+ requests/sec.
- Developed page-locked CUDA memory pools (cudaMallocHost) and IPC mechanisms to reduce page-fault overhead and accelerate host-to-device transfers.
- Built containerized FastAPI REST/gRPC gateways; automated benchmarks measured RPS, VRAM, cosine similarity, and perplexity.

Multimodal Vision RAG Engine | Distributed AI Search Pipeline **July 2026**
Python, PyTorch, CLIP, BM25, PyMuPDF, Qdrant, FastAPI, vLLM, Celery, Redis, Next.js

- Developed an end-to-end multimodal GenAI pipeline combining Large Vision Models (LLaVA / Qwen-VL) and CLIP embeddings for automated visual question answering and multi-page document parsing.
- Designed hybrid retrieval (Qdrant dense vectors + BM25 keyword matching) with Reciprocal Rank Fusion (RRF, k=60), reducing context retrieval latency by 42% to 38.4 ms.
- Implemented sub-5 ms LRU/Redis query-vector caching and FastAPI WebSocket streaming to maintain sub-200 ms time-to-first-token (TTFT) for client applications.
- Engineered a distributed asynchronous task queue using Celery, Redis, and PyMuPDF for concurrent multi-page document processing, reducing end-to-end ingestion time by 38%.

Achievements / Certifications

- Active competitive programmer with peak ratings of 1801 on CodeChef and 1787 on LeetCode.
- Achieved an All India Rank (AIR) of 8741 in JEE Mains 2023 (Top 0.8% out of 1.2M+ candidates).
- Google Cloud Gen AI Academy APAC Edition Certificate, Google Cybersecurity Professional Certificate, and IBM Cybersecurity Fundamentals.

Technical Skills

ML Infra & Systems: CUDA (Basics), Docker, Triton Inference Server, gRPC, Multiprocessing, Linux CLI
AI/ML & GenAI: PyTorch, TensorFlow, HuggingFace, LLMs, LVMs, TensorRT, vLLM, RAG Pipelines, Model Quantization
Languages: Python, C/C++, JavaScript (ES6+), SQL, Java
Frameworks & Databases: FastAPI, Flask, Next.js, PostgreSQL, Redis, SQLite, RESTful API Design
Developer Tools: Git/GitHub, TensorBoard, Model Evaluation & Benchmarking Tools

Leadership / Extracurricular

Student Activity Council (Cultural) & Sports

- **Event Management:** Coordinated the IPL Auction and Pictionary events for Aurora (annual cultural fest) and spearheaded 10+ Orientation Week events, improving student satisfaction scores by 15%.
- **Athletics:** Karate Black Belt; 9+ years; 30+ tournament appearances; Gold/Silver/Bronze medalist; competitive swimmer.